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Inventor(s)

Agrawal; Mahi et al.

System and Method to Control Merger Belt Slip

Abstract

The present disclosure is a system for controlling an agricultural machine. The system has a merger belt assembly comprising a merger belt, a controller configured to selectively control the merger belt assembly, and a user input configured to provide a desired belt speed command to the controller indicating a desired operating speed of the merger belt. The controller provides instructions to the merger belt assembly of the agricultural machine to operate the merger belt at the desired operating speed. Further, the controller is configured to monitor an actual merger belt speed of the merger belt assembly and provide a response when the actual merger belt speed is outside of a desired operating speed threshold established by the desired belt speed command.

Inventors: Agrawal; Mahi (Pune, IN), Gorivale; Sameer (Pune, IN)

Applicant: Deere & Company (Moline, IL)

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Background/Summary

FIELD OF DISCLOSURE

[0001] The present disclosure relates to an agricultural vehicle having one or more conveyor and more specifically to selectively controlling the one or more conveyor based on operating conditions.

BACKGROUND OF THE DISCLOSURE

[0002] Known agricultural machines, such as windrowers, often have a merger belt configured to redirect harvested material away from a path of travel of the windrower. The speed of the merger belt is typically established by the user to reposition the harvested material to the desired location. However, if the user improperly sets the merger belt speed for the operating conditions of the agricultural machine, the harvested material may clog the agricultural machine, among other things.

SUMMARY

[0003] One embodiment is a system for controlling an agricultural machine. The system has a merger belt assembly comprising a merger belt, a controller configured to selectively control the merger belt assembly, and a user input configured to provide a desired belt speed command to the controller indicating a desired operating speed of the merger belt. The controller provides instructions to the merger belt assembly of the agricultural machine to operate the merger belt at the desired operating speed. Further, the controller is configured to monitor an actual merger belt speed of the merger belt assembly and provide a response when the actual merger belt speed is outside of a desired operating speed threshold established by the desired belt speed command.

[0004] In one example of this embodiment, the controller is configured to provide one or more of decreasing a header speed, increasing a merger belt speed, reducing a vehicle speed, and adjusting a roll gap as the response. In part of this example, the controller is configured to automatically execute the response.

[0005] In yet another example of this embodiment, the controller is configured to provide a recommended belt speed to a user via a display. In part of this example, the controller is configured to determine the recommended belt speed based on operating conditions of the agricultural machine. In one aspect of this part, the controller is configured to consider one or more of a header speed, a vehicle speed, and a roll gap as operating conditions of the agricultural machine.

[0006] In yet another example of this embodiment the controller is configured to automatically adjust the desired operating speed as part of the response. In part of this example, the controller is configured to automatically adjust the desired operating speed based on operating conditions of the agricultural machine. In one aspect of this part, the controller is configured to consider one or more of a header speed, a belt slip, a vehicle speed, and a roll gap for the operating conditions.

[0007] In another example of this embodiment, the controller is configured to determine the desired belt speed command from data from the agricultural machine, the data comprising one or more of a header speed, a throw distance, and a header load. In part of this example, the data comprises the throw distance and the throw distance is input by a user through the user input.

[0008] Yet another embodiment of this disclosure is a method for managing a merger belt speed of a merger belt assembly on an agricultural machine. This method includes evaluating operating conditions of the agricultural machine with a controller, determining a desired belt speed of a belt of the merger belt assembly with the controller based on the operating conditions, instructing the merger belt assembly to operate the belt at the desired belt speed, monitoring an actual belt speed of the belt of the merger belt assembly with the controller through a sensor, identifying when the actual belt speed is not within a threshold established by the desired belt speed with the controller, and generating an automatic response on the agricultural machine with the controller when the desired belt speed is not within the threshold.

[0009] In one example of this embodiment the automatic response is one or more of modifying a header speed, reducing a vehicle speed, and adjusting a roll gap. In another example, the evaluating

the operating conditions includes considering one or more of a header speed, a throw distance, and a header load. In yet another example, the evaluating the operating conditions includes considering at least one input provided from a user input configured to identify a user-preferred speed of the belt of the merger belt assembly.

[0010] In yet another example of this embodiment the desired belt speed is a user-selected belt speed selected by a user through a user-interface. In part of this example, the user-selected belt speed is a recommended belt speed generated by the controller based on operating conditions.

[0011] In yet another example of this embodiment the automatic response is to shut down the merger belt assembly when the actual belt speed is within a shutdown threshold established by the desired belt speed with the controller.

[0012] Yet another embodiment of this disclosure is a method for controlling a merger belt assembly of an agricultural machine. The method includes monitoring operating conditions of the agricultural machine with a controller, setting a desired belt speed of a merger belt of the merger belt assembly based on the operating conditions, comparing an actual belt speed of the merger belt assembly with the desired belt speed, and providing an automatic response when the actual belt speed is outside of a belt speed threshold established by the desired belt speed.

[0013] In one example of this embodiment the automatic response includes modifying one or more of the operating conditions.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above-mentioned aspects of the present disclosure and the manner of obtaining them will become more apparent and the disclosure itself will be better understood by reference to the following description of the embodiments of the disclosure, taken in conjunction with the accompanying drawings, wherein:

[0015] FIG. 1 is a perspective view of a vehicle, according to one embodiment;

[0016] FIG. 2 is a perspective view of a vehicle including a merger device, according to one embodiment;

[0017] FIG. 3 is a perspective view of a vehicle including a merger device, according to one embodiment;

[0018] FIG. 4 is a perspective view of a cab of a vehicle, according to one embodiment;

[0019] FIG. 5 is a schematic view of a control system, according to one embodiment; and

[0020] FIG. 6 is a method flow chart for controlling an vehicle, according to one embodiment.

[0021] Corresponding reference numerals are used to indicate corresponding parts throughout the several views.

DETAILED DESCRIPTION

[0022] The embodiments of the present disclosure described below are not intended to be exhaustive or to limit the disclosure to the precise forms in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may appreciate and understand the principles and practices of the present disclosure.

[0023] Referring to FIG. 1, an agricultural vehicle **100** is illustrated. The agricultural vehicle **100** may include an automated repositioning system having any one or more of the controllers, sensors, and devices described herein. Although a self-propelled windrower having a dedicated power source **108** is shown in FIG. 1, the present disclosure also considers other self-propelled and pull-type agricultural harvesting equipment, such as cutters, mowers, mower-conditioners, forage harvesters, and combines. The vehicle **100** may include an operator compartment or cab **102** where an operator may direct or control the operation of the vehicle **100**. The vehicle **100** may include a frame **104** to which one or more ground engaging apparatus **106**, such as wheels or tracks, are

operably connected. The vehicle **100** may include a power source and a transmission operably coupled to one or more ground engaging apparatus **106**. In one aspect of this disclosure, a vehicle speed control assembly **204** may be selectively controlled by a controller **202** to selectively alter the vehicle speed by automatically managing the power source, transmission, and ground engaging apparatus **106**.

[0024] The vehicle **100** may include a harvesting device or attachment, such as a harvesting header **120**, which can harvests a variety of crops including, but not limited to, hay, corn, and soybeans among other things. The harvesting header **120** can be a rotary header or a draper header controlled by a header assembly **206**. The header assembly **206** may be selectively controllable by the controller **202** to alter the speed of either a rotary header or a draper header. Further, the vehicle **100** may include a single harvesting header **120** or multiple harvesting headers **120**. The harvesting header **120** may include a lowered or operating position, as shown, and a raised or non-operating position. The harvesting header **120** may include an active or operating condition and an inactive or non-operating condition. In the operating condition, the components of the harvesting header **120** are moving or otherwise operational and selectively controlled by the controller **202** through the header assembly **206**.

[0025] The harvesting header **120** may direct harvested crop through a roll gap assembly **210**. The roll gap assembly **210** may be selectively moveable to alter a gap for crop to pass there through. More specifically, the roll gap assembly **210** may include a conditioning roller and an opposing member that are configured to cooperatively condition a crop material that passes through a gap between the conditioning roller and the opposing member. This gap may be alterable through one or more actuator selectively controlled by the controller **202**. The gap provided by the roll gap assembly **210** may be selectively defined through a user selection or automatically controlled with the controller **202**. See at least U.S. Pat. No. 10,806,078 for an example of a system for adjusting a gap for conditioning rollers of the harvesting header **120**, the contents of which are incorporated herein in entirety.

[0026] Referring now to FIG. 2, the agricultural vehicle **100** may include another harvesting device or attachment, such as a merger device **122** having one or more conveyors **126**, **128**. The merger device **122** may be part of a merger belt assembly **208** and be selectively controlled by the controller **202**. The merger device **122** may include a lowered or other operating position and a raised or other non-operating position. The merger device **122** may include an active or operating condition and an inactive or non-operating condition selectively implemented by controller **202** through the merger belt assembly **208**. In the operating condition, the one or more conveyors **126**, **128** are moving or otherwise operational. The merger device **122** may include a conveyor **126** which receives harvested crop from the harvesting header **120** and delivers the crop to a cross-conveyor **128**. The cross-conveyor **128** receives the crop from the conveyor **126** and delivers the crop to either side of the vehicle **100**. Additionally or alternatively, the vehicle **100** could include one or more merger devices **122** positioned on or near the rear of the vehicle **100**.

[0027] The merger belt assembly **208** may have a power source or motor **150** selectively controlled by the controller **202** to move the cross-conveyor or merger belt **128** to distribute crop from the conveyor **126** out one side of the agricultural vehicle **100**. The motor **150** may be hydraulic, pneumatic, or electrically powered and is selectively powered by the controller **202** to rotate the merger belt **128** thereabout. In one aspect of this disclosure, the speed and angle of the merger belt **128** may be adjustable such that a throw distance can be modified by the merger belt assembly **208**. The throw distance may be the distance harvested material is thrown from the merger belt **128** before it contacts the underlying surface. In view thereof, in one aspect of this disclosure the user may input a desired throw distance through the user input **212** and the controller **202** can adjust one or more of the speed of the merger belt **128** or the angle of the merger belt to accommodate the user-input throw distance.

[0028] Referring now to FIG. 3, the merger device **122** may include a deflector **124** which can

direct the crop after it leaves the cross-conveyor **128**. The deflector **124** may include a lowered or operating position, as shown, and a raised or non-operating position. In the lowered position, the deflector **124** engages or directs the crop after it leaves the cross-conveyor **128**. In the raised position, the deflector **124** remains disengaged from the flow of crop as it leaves the cross-conveyor **128**.

[0029] Referring now to FIG. **4**, an interior of a cab **102** of the agricultural vehicle **100** is illustrated. The cab **102** may include one or more operator controls **130**, such as a direction input device **132**, a device selector **134**, a user input **212**, and a display **214**. The direction input device **132** provides an interface for an operator to control the direction of the vehicle **100**. The device selector **134** provides an interface for an operator to select or pre-select which devices are to be controlled. Similarly, the user input **212** may be a touch screen or other input device wherein the user can selectively control any of the assemblies **204**, **206**, **208**, **210** discussed herein. Further, the display **214** may provide a visual indicator to the operator about the operating conditions of any of the assemblies **204**, **206**, **208**, **210** of the agricultural vehicle **100** among other things.

[0030] Referring now to FIG. **5**, a schematic representation of a control system **200** is illustrated. The control system **200** may selectively control any of the assemblies of the agricultural vehicle **100**. More specifically, the controller **202** may be one or more processor with access to a memory unit to store data and execute commands, among other things. The controller **202** may be a standard controller on the agricultural vehicle **100** used to execute common functions for the agricultural vehicle **100**. Alternatively, the controller **202** can be a dedicated hardware component for the merger belt assembly **208**. In this embodiment, the controller **202** may communicate with one or more other controllers of the agricultural vehicle **100** to implement the teachings discussed herein. In yet another embodiment, the controller **202** may be part of a remote device that communicates wirelessly with the agricultural vehicle **100** to implement the teachings discussed herein. A person skilled in the art understands many different hardware components could be used to implement the methods discussed herein, and this disclosure contemplates using any known hardware or technical methodology for implementing the teachings discussed herein on the agricultural vehicle **100**.

[0031] The controller **202** may communicate with the vehicle speed control assembly **204** to selectively alter the speed of the agricultural vehicle **100**. More specifically, one or more vehicle speed sensor **216** may communicate a vehicle speed to the controller **202**. In one embodiment of this disclosure, the controller **202** may selectively manipulate the vehicle speed control assembly **204** to alter the speed of the agricultural vehicle **100**. For example, the controller **202** may monitor the vehicle speed sensor **216** alter the vehicle speed with the vehicle speed control assembly **204** to change the vehicle speed to a desired vehicle speed based on the working conditions. In one aspect of this disclosure, the controller **202** may control the vehicle speed control assembly **204** by selectively altering one or more of the power source, transmission, and engaging apparatus **106** of the agricultural vehicle **100** to alter the vehicle speed.

[0032] Similarly, the controller **202** may communicate with the header assembly **206** to selectively control one or more components of the harvesting header **120**. In the example where the harvesting header **120** is a rotary header, the controller **202** may communicate with a header speed sensor **218** to determine the speed at which the rotary header is rotating. Similarly, the controller **202** may communicate with a header load sensor **220** to determine the load of crop being processed by the harvesting header **120**. In one aspect of this disclosure, the controller **202** may selectively alter the operating parameters of the header assembly **206**. More specifically, if the header assembly **206** is a rotary header, the controller **202** may selectively alter the rotational speed of the rotors by monitoring the header speed sensor **218** and selectively changing the header speed based on the operating conditions of the agricultural vehicle **100**. Similarly, the controller **202** may also monitor the header load sensor **220** to selectively alter things like the speed of conveyor **126** or the speed of a draper assembly if the harvesting header **120** is a draper header. Accordingly, in one aspect of this disclosure the controller **202** can monitor and selectively alter the operating parameters of the

header assembly **206**.

[0033] The controller **202** may also communicate with, and selectively control, the merger belt assembly **208**. More specifically, the controller **202** may communicate with a merger belt sensor **222** to identify the operating speed of the merger belt **128**. Further, the controller **202** may selectively control the operating speed of the merger belt **128** by altering the speed of the motor **150** of the merger device **122**. In view thereof, the controller **202** may both instruct a desired speed of the merger belt **128** through the motor **150** and monitor the actual speed of the merger belt **128** with the merger belt sensor **222**.

[0034] The controller **202** may also selectively alter the roll gap assembly **210** to provide an ideal roll gap for the operating conditions. In this example, the controller **202** may selectively enlarge, or narrow, the roll gap with the roll gap assembly **210** so the agricultural vehicle **100** is operating under ideal conditions.

[0035] The controller **202** may also be in communication with the display **214** to convey to the user any known information about the operating conditions of any of the assemblies **204**, **206**, **208**, **210** among other things. In one example, the display **214** may be a touchscreen display in the cab **102** of the agricultural vehicle **100**. Alternatively, the display **214** may not be a touchscreen but rather may be a screen or light or combination of lights configured to provide a visual message to a user in the cab **102**. In yet another embodiment contemplated herein, the display **214** may be a wireless device that communicates wirelessly with the controller **202**. More specifically, the display **214** may be on a wireless computer such as a laptop, tablet, smartphone, or other similar electronic device.

[0036] While the term “display” is used herein for display **214**, this disclosure contemplates conveying messages to a user with non-visual means as well. For example, the display **214** may have one or more of audio and haptic feedback for the user as well. In one example, the display **214** only includes one or more of audio and haptic feedback and does not provide a visual message to the user. In other embodiments, the display **214** provides audio, visual, and haptic feedback to the user as instructed from the controller **202**. Regardless, the controller **202** can selectively engage the display **214** to provide a message to the user.

[0037] The controller **202** may also communicate with the user input **212** to provide desired functions for one or more of the assemblies **204**, **206**, **208**, **210**, among other things. For example, the user input **212** can provide an input to the controller **202** indicating a desired vehicle speed, header speed, merger belt speed, merger throw distance, and roll gap distance among other things. The controller **202** may communicate with the user input **212** such that the controller **202** can manipulate one or more of the assemblies **204**, **206**, **208**, **210** responsive to the desired operating parameters input by a user through the user input **212**. In one aspect of this disclosure, the display **214** and user input **212** may be the same touchscreen device. Similarly, the user input **212** may be a wireless device that communicates wirelessly with the controller **202**. More specifically, the user input **212** may be a wireless computer such as a laptop, tablet, smartphone, or other similar electronic device.

[0038] Referring now to FIG. **6**, one embodiment of the present disclosure is illustrated. More specifically, FIG. **6** illustrates a method for controlling an agricultural machine having a merger device **600**. The method **600** may be implemented by the controller **202** or one or more controller of the agricultural vehicle **100**. Alternatively, a remote device in wireless communication with the agricultural vehicle may implement the method **600** discussed herein. The method **600** may be implemented when the agricultural vehicle **100** is in the operational configuration preparing to, or already engaging, the header assembly **206** to harvest crop. In one aspect of this disclosure, the method **600** may commence when a user initiates the method through the user input **212**. Alternatively, the method **600** may begin automatically when the agricultural machine **100** is in the operational configuration.

[0039] Regardless, in box **602** the controller **202**, or an equivalent thereof, may establish a desired

belt speed of the merger belt **128** of the merger belt assembly **208**. The desired belt speed may be input by the user via the user input **212** or automatically determined by the controller **202** or an equivalent thereof based on the operating conditions of one or more of the assemblies **204**, **206**, **208**, **210**. When the desired belt speed is established via the user input **212**, the user may engage the user input **212** to set the desired belt speed of the merger belt **128**. The user input **212** may provide selectable speeds for the merger belt **128** that the user can select to establish the desired belt speed or the user may input a desired speed within an available speed range (i.e., the desired revolutions per minute of the merger belt **128** within the functional range of the merger belt assembly **208**).

[0040] The controller **202** may be configured to send a signal to the motor **150** to operate the motor **150** at the desired belt speed. In one example, the controller **202** may have predefined therein the operating parameters of the motor **150** such that the controller **202** may provide a corresponding signal to the motor **150** or a system thereof to operate the motor **150** at the desired speed under normal operating conditions. That is to say, the controller **202** may have predefined operating parameters for the motor **150** that would provide the corresponding desired speed of the merger belt **128** under normal operating conditions when the merger belt **128** is not experiencing increased or unexpected loads. In one example, the motor **150** is a hydraulic or pneumatic motor and the controller **202** may operate one or more valves of an electro-hydraulic or electro-pneumatic system to provide a fluid pressure and flow to the motor **150** to produce the desired speed under normal working conditions. Similarly, if the motor **150** is an electric motor the controller **202** may power the electric motor with sufficient electrical power to generate the desired speed under normal operating conditions.

[0041] In one aspect of this disclosure, the controller **202** or an equivalent thereof may provide a recommended belt speed as an option on the user input **212** or display **214** as part of box **602**. In this example, the recommended belt speed may be determined by the controller **202** or an equivalent thereof by monitoring or otherwise identifying the operating conditions of one or more of the assemblies **204**, **206**, **208**, **210**. More specifically, the controller **202** or an equivalent thereof may identify one or more operating condition such as the vehicle speed from the vehicle speed sensor **216**, the header speed from the header speed sensor **218**, the header load from the header load sensor **220**, the merger belt speed from the merger belt sensor **222**, the roll gap from the roll gap assembly **210**, and a desired throw distance from the user input **212** among other operating conditions. In this example, the controller **202** or an equivalent thereof may provide a recommended desired belt speed via the display **214** or the user input **212** based on one or more of the operating conditions discussed herein.

[0042] In another aspect of this disclosure, the user may input anticipated or desired operating conditions with the user input **212** to allow the controller **202** or an equivalent thereof to generate a recommended desired belt speed in box **602**. More specifically, the user may input via the user input **212** or other similar device the expected or desired operating conditions of the agricultural vehicle **100**. The expected or desired operating conditions may include the vehicle speed, header speed, header load, roll gap, and throw distance among other things. The user may input values for one or more of these operating conditions and the controller **202** or an equivalent thereof may generate a recommended desired belt speed in box **602**.

[0043] The desired belt speed may also be determined automatically by the controller **202** or an equivalent thereof. In this example, the desired belt speed may be established based on the operating conditions of the other assemblies **204**, **206**, **208**, **210** among other things. For example, the desired belt speed may be established by the controller **202** or an equivalent thereof by considering one or more of the speed of the agricultural vehicle **100**, the header speed or header load of the header assembly **206**, and the roll gap of the roll gap assembly **210** among other things. In one example, the controller **202** or an equivalent thereof may have a lookup table that correlates a desired belt speed of the merger belt **128** with one or more of the operating conditions of the

assemblies **204, 206, 208, 210**. Alternatively, the controller **202** or an equivalent thereof may use an algorithm or other equation to determine the desired belt speed of the merger belt **128** in box **602**. Regardless, in box **602** the controller **202** or an equivalent thereof may establish a desired belt speed either through a user input or an automatic process based on information from the other assemblies **204, 206, 208, 210**.

[0044] In box **604**, the controller **202** or an equivalent thereof may instruct the agricultural machine **100** to operate the merger assembly **206** at the desired belt speed. More specifically, in one example the controller **202** may instruct a hydro-electric system to rotate the motor **150** such that the merger belt **128** rotates at the desired belt speed established in box **602**. Similarly, in another example the controller **202** may instruct an electrical system utilizing electric motors to rotate the motor **150** such that the merger belt **128** rotates at the desired belt speed established in box **602**. Further still, in another embodiment the controller **202** may instruct an electro pneumatic system to rotate the motor **150** such that the merger belt **128** rotates at the desired belt speed established in box **602**. Regardless of the system type, in box **604** the controller **202** or an equivalent thereof instructs the merger belt assembly **208** to rotate the merger belt **128** at the desired belt speed from box **602**.

[0045] After the controller **202** or an equivalent thereof has engaged the merger belt assembly **208** to rotate the merger belt **128** at the desired speed in box **604**, the controller **202** or an equivalent thereof may monitor the merger belt sensor **222** to identify an actual belt speed of the merger belt **128** in box **606**. The actual belt speed may be the speed at which the merger belt **128** is actually rotating in the merger belt assembly **208**. In other words, the actual belt speed may be the measured speed of the merger belt **128** as observed by the merger belt sensor **222**.

[0046] In box **608**, the controller **202** or an equivalent thereof can compare the actual merger belt speed determined in box **606** with the desired belt speed established in box **602** to determine whether the amount of belt slip that is occurring in the merger belt assembly **208**. As used herein, “belt slip” may refer to the difference between the actual merger belt speed and the desired belt speed. In one aspect of this disclosure, belt slip may be identified as a percentage variance of the actual belt speed compared to the desired belt speed. In one example, the threshold of box **608** may be any actual belt speed that is within ten percent of the desired belt speed from box **602**.

Alternatively, the threshold of box **608** may be preset values that are associated with user-selectable desired belt speeds from box **602**.

[0047] While the threshold of box **608** is described herein in one example as ten percent of the desired belt speed, other percentages are contemplated herein as well. For example, in another embodiment the threshold percentage may be more than ten percent the desired belt speed. Further still, in other embodiments the threshold percentage may be less than ten percent the desired belt speed. Accordingly, this disclosure contemplates utilizing any threshold that may be reasonable for an agricultural machine **100** having a merger belt assembly **208**.

[0048] In another aspect of this disclosure, the threshold value used in box **608** may be a user-selectable value. For example, the user may use the user input **212** to communicate a user-desired threshold value to be applied in box **608**. The controller **202** may store the user-desired threshold value in a memory unit and use the user-desired threshold value in box **608**. Accordingly, one aspect of this disclosure allows a user to establish the threshold value applied in box **608**.

[0049] If the actual merger belt speed is within the threshold in box **608**, the controller **202** or an equivalent thereof may continue to execute boxes **602, 604, 606** to ensure the merger belt assembly **208** continues operating as expected within the threshold of the desired belt speed.

[0050] If the actual merger belt speed is not within the threshold of the desired belt speed in box **608**, the controller **202** or an equivalent thereof may check if the actual merger belt speed is within a shutdown threshold in box **610**. The shutdown threshold may be based on the desired belt speed established in box **602** and be such that when the actual speed is within the shutdown threshold it is an indication that the merger belt assembly **208** is being overloaded and unable to move the merger belt **128** at a sufficient speed to continue. In one example, the shutdown threshold may be a speed

that is only ten percent or less of the desired belt speed established in box **602**. However, this disclosure also considers shutdown thresholds that are greater than ten percent the desired belt speed and shutdown thresholds that are less than ten percent the desired belt speed.

[0051] Regardless, if the actual merger belt speed is within the shutdown threshold in box **610**, the controller **202** or an equivalent thereof may shut down the merger belt assembly **208** and quit directing power to the motor **150** powering the belt **128** in box **614**. In addition to shutting down the merger belt assembly **208**, other embodiments of this disclosure contemplate altering any of the other assemblies **204**, **206**, **210** responsive to the actual merger belt speed being in the shutdown threshold in box **610**. Accordingly, in one aspect of this disclosure the controller **202** or an equivalent thereof may automatically slow or stop any of the assemblies **204**, **206**, **208**, **210** discussed herein when the actual merger belt speed is within the shutdown threshold in box **610**. [0052] While box **610** is discussed herein to compare the actual merger belt speed to a shutdown threshold, other embodiments considered herein may omit this step. Accordingly, box **610** is only one example of a step contemplated by this disclosure and may not be included in alternative embodiments.

[0053] If the actual merger belt speed is not within the threshold of the desired belt speed in box **608**, or the shutdown threshold of box **610** when utilized, the controller **202** or an equivalent thereof may provide a response in box **612**. In one aspect of this disclosure, the response from box **612** may be a visual indication provided on the display **214** indicating to the user that the actual merger belt speed measured by the merger belt sensor **222** is not within the thresholds of box **608**, or **610** when utilized. The visual indication may be in the form of an illuminated light or a graphic that indicates to the user that the actual merger belt speed is not within the threshold of the desired belt speed. Alternatively, the response of box **612** may be an audible noise or haptic feedback that can be perceived by the user to indicate the actual merger belt speed is not within the threshold of the desired belt speed.

[0054] In another aspect of this disclosure, the response of box **612** may be modifying one or more operating condition of one or more of the assemblies **204**, **206**, **208**, **210**. In one example, the response of box **612** may be automatically controlling the vehicle speed control assembly **204** to slow the agricultural vehicle **100** with the controller **202** or an equivalent thereof when the actual merger belt speed is not within the threshold of the desired merger belt speed. In another example, the response of box **612** may include automatically controlling the header assembly **206** to slow the header speed or draper speed with the controller **202** or an equivalent thereof when the actual merger belt speed is not within the threshold of the desired merger belt speed. In another example, the response of box **612** may include automatically controlling the merger belt assembly **208** to increase the merger belt speed with the controller **202** or an equivalent thereof when the actual merger belt speed is not within the threshold of the desired merger belt speed. In yet another example, the response of box **612** may include automatically controlling the roll gap assembly **210** to adjust the roll gap with the controller **202** or an equivalent thereof when the actual merger belt speed is not within the threshold of the desired merger belt speed. In yet another example of this disclosure, the response of box **612** may include modifying more than one of the assemblies **204**, **206**, **208**, **210** with the controller **202** or an equivalent thereof when the actual merger belt speed is not within the threshold of the desired merger belt speed.

[0055] In use, one embodiment of this disclosure allows the user to input the desired belt speed of the merger belt assembly **208** via the user input **212** before or while initiating a harvesting operation of the agricultural vehicle **100**. The controller **202** will then instruct the motor **150** of the merger belt assembly **208** to rotate the merger belt **128** at the user-input desired belt speed. The controller **202** will monitor the actual belt speed of the merger belt **128** with the merger belt sensor **222**. If the actual speed of the merger belt falls outside of a threshold of the desired merger belt speed (i.e., more than ten percent variance between actual merger belt speed and the desired merger belt speed), the controller **202** may automatically provide a response. The response may include

one or more of providing an indication of the issue on the display **214**, modifying the vehicle speed with the vehicle speed control assembly **204**, modifying the operating conditions of the header assembly **206**, modifying the conditions of the merger belt assembly **208**, and adjusting the gap of the roll gap assembly **210**.

[0056] In yet another use embodiment, the controller **202** may present one or more efficient set point values for the desired merger belt speed to the user through the display **214** or the user input **212**. The one or more efficient set point values are presented to the user for the user to select the desired belt speed from those presented. In this example, the controller **202** may consider the current operating conditions or user-input operating conditions of the agricultural machine **100** when the controller **202** presents the one or more efficient set point values. For example, the controller **202** may consider one or more of the gap of the roll gap assembly **210**, the user-selected throw distance of the merger belt assembly **208**, the speed or expected load of the header assembly **206**, and the desired or current vehicle speed when determining the one or more efficient set point values for the desired merger belt speed. Then, the user may select one of the presented efficient set point values for the desired merger belt speed. The controller **202** may then execute the method **600** discussed herein based on the user selected desired belt speed.

[0057] In another use embodiment, the user may set the desired merger belt speed to be automatically determined by the controller **202**. In this embodiment, the controller **202** may set the desired merger belt speed to any speed that will provide the desired results based on the operating conditions. For example, if the agricultural vehicle **100** increases the speed of travel, the controller **202** may automatically increase the desired merger belt speed correspondingly. Similarly, if the controller **202** identifies that the actual merger belt speed is slowing relative to the desired merger belt speed, the controller **202** may increase the desired belt speed to increase the amount of harvested material that can be processed by the merger belt assembly **208**. Similarly, the controller **202** may automatically adjust the desired merger belt speed if the gap of the roll gap assembly **210** changes. In one aspect of this embodiment, the controller **202** may also automatically adjust any of the assemblies **204**, **206**, **208**, **210** when the operating conditions would result in the actual belt speed falling outside of the threshold of the desired belt speed. Among other things, the controller **202** could automatically do one or more of decrease header speed, increase merger belt speed, reduce vehicle speed, and adjust the roll gap.

[0058] While embodiments incorporating the principles of the present disclosure have been described hereinabove, the present disclosure is not limited to the described embodiments. Instead, this application is intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A system for controlling an agricultural machine, comprising: a merger belt assembly comprising a merger belt; a controller configured to selectively control the merger belt assembly; and a user input configured to provide a desired belt speed command to the controller indicating a desired operating speed of the merger belt; wherein the controller is configured to provide instructions to the merger belt assembly of the agricultural machine to operate the merger belt at the desired operating speed; wherein the controller is configured to monitor an actual merger belt speed of the merger belt assembly and provide a response when the actual merger belt speed is outside of a desired operating speed threshold established by the desired belt speed command.
2. The system of claim 1, wherein the controller is configured to provide one or more of decreasing a header speed, increasing a merger belt speed, reducing a vehicle speed, and adjusting a roll gap as the response.

3. The system of claim 2, wherein the controller is configured to automatically execute the response.
4. The system of claim 1, wherein the controller is configured to provide a recommended belt speed to a user via a display.
5. The system of claim 4, wherein the controller is configured to determine the recommended belt speed based on operating conditions of the agricultural machine.
6. The system of claim 5, wherein the controller is configured to consider one or more of a header speed, a vehicle speed, and a roll gap as operating conditions of the agricultural machine.
7. The system of claim 1, wherein the controller is configured to automatically adjust the desired operating speed as part of the response.
8. The system of claim 7, wherein the controller is configured to automatically adjust the desired operating speed based on operating conditions of the agricultural machine.
9. The system of claim 8, wherein the controller is configured to consider one or more of a header speed, a belt slip, a vehicle speed, and a roll gap for the operating conditions.
10. The system of claim 1, wherein the controller is configured to determine the desired belt speed command from data from the agricultural machine, the data comprising one or more of a header speed, a throw distance, and a header load.
11. The system of claim 10, wherein the data comprises the throw distance and the throw distance is input by a user through the user input.
12. A method for managing a merger belt speed of a merger belt assembly on an agricultural machine, the method comprising: evaluating operating conditions of the agricultural machine with a controller; determining a desired belt speed of a belt of the merger belt assembly with the controller based on the operating conditions; instructing the merger belt assembly to operate the belt at the desired belt speed with the controller; monitoring an actual belt speed of the belt of the merger belt assembly with the controller through a sensor; identifying when the actual belt speed is not within a threshold established by the desired belt speed with the controller; and generating an automatic response on the agricultural machine with the controller when the desired belt speed is not within the threshold.
13. The method of claim 12, wherein the automatic response is one or more of modifying a header speed, reducing a vehicle speed, and adjusting a roll gap.
14. The method of claim 12, wherein evaluating the operating conditions comprises considering one or more of a header speed, a throw distance, and a header load.
15. The method of claim 12, wherein evaluating the operating conditions comprises considering at least one input provided from a user input configured to identify a desired speed of the belt of the merger belt assembly.
16. The method of claim 12, wherein the desired belt speed is a user-selected belt speed selected by a user through a user-interface.
17. The method of claim 16, wherein the user-selected belt speed is a recommended belt speed generated by the controller based on operating conditions.
18. The method of claim 12, wherein the automatic response is to shut down the merger belt assembly when the actual belt speed is not within a shutdown threshold established by the desired belt speed with the controller.
19. A method for controlling a merger belt assembly of an agricultural machine, comprising: monitoring operating conditions of the agricultural machine with a controller; setting a desired belt speed of a merger belt of the merger belt assembly based on the operating conditions; comparing an actual belt speed of the merger belt assembly with the desired belt speed; and providing an automatic response when the actual belt speed is outside of a belt speed threshold established by the desired belt speed.
20. The method of claim 19, wherein the automatic response comprises modifying one or more of the operating conditions.

